

Canonical Determinant Cells in a Primitive Möbius Tail:

Support–Cardinality Bounds, All–Content Cell Pruning, and the Residue–Cell Zero–Mode Gate

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Abstract

Let

$$N_m(j) = mj + h, \quad N_n(j) = nj + h$$

be two positive affine targets in the primitive opened-row packet of the preceding Möbius-tail reduction. The row and orbit scales are $Q = X^{\beta+o(1)}$ and $J = X^{1-\beta+o(1)}$, with $QJ \asymp X$. For each off-diagonal row–orbit triple define the full target content and the canonical normalized determinant by

$$c = (N_m(j), N_n(j)), \quad \Delta^\# = \frac{m-n}{c}.$$

The first quantity is factorization-independent, and the primitive target identity makes $\Delta^\#$ an integer. If $U = N_m(j)/c$ and $V = N_n(j)/c$, then

$$(U, V) = 1, \quad mV - nU = h\Delta^\#.$$

Thus $\Delta^\#$ is the determinant of the canonically reduced target pair, rather than the determinant attached to one selected divisor factorization.

For exact content c and an allowed set $\Omega_c \subset \mathbb{Z}$, let

$$\mathcal{H}_c(\Omega_c) = \#\{\delta \in \Omega_c : |\delta| \ll Q/c\}.$$

The difference graph $m - n = c\delta$ has both degrees at most $\mathcal{H}_c(\Omega_c)$. A Schur estimate, the physical row ℓ^2 bound, and the single orbit class modulo c give, for each of the three retained raw channels,

$$|\mathcal{S}(c \leq Z, \Delta^\# \in \Omega_c)| \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon Q \log(2L) \sum_{c \leq Z} \left(1 + \frac{J}{c}\right) \mathcal{H}_c(\Omega_c).$$

This is an unconditional capacity bound for every canonical determinant spectrum. It gives power saving whenever the displayed orbit-weighted support count is polynomially smaller than JQ ; the explicit cases below include translated windows far from the row diagonal and individual normalized-determinant residue cells.

Splicing this estimate with the large-target-content theorem of the preceding paper removes the auxiliary restriction $c \leq Z$. If I is any translated interval of length W and $a \pmod{q}$ is one residue class, then for any splitting threshold $1 \leq C \ll Q$ the full physical cell satisfies

$$|\mathcal{S}(\Delta^\# \in I, \Delta^\# \equiv a \pmod{q})| \ll X^\varepsilon XQ \left[J^{-1} + C^{-1} + \left(1 + \frac{W}{q}\right) \left(\frac{C}{X} + \frac{1}{Q}\right) \right].$$

Consequently, if $1 + W/q \leq X^{s+o(1)}$ with $s \geq 0$ and $s < \beta$, every fixed saving exponent below

$$\min\{1 - \beta, \beta - s\}$$

is available after optimizing C . For one residue class over the entire determinant range, the relaxed saving is $\min\{1 - \beta, \beta, \rho\}$ when $q = X^{\rho+o(1)}$.

At $\beta = 267/400$, an arbitrary translated J -window or a single class modulo $q \asymp J$ has every fixed saving below $133/400$ after the all-content splice. If the J -window is centered at $|\Delta^\#| \asymp Q$, its content is automatically bounded and the margin improves to $67/200$. A translated window of width $X^{1/2+o(1)}$ has every fixed saving below $67/400$.

The result is cellwise. Summing all determinant cells absolutely recovers the original row family, and an exact finite Fourier formula shows that the unweighted reassembly is the $r = 0$ coefficient of an auxiliary residue-cell transform. Nonzero cell energy alone therefore does not close the residual. A candidate next route must restore the prime–Möbius core while retaining the fixed residue factors, pair mask, and matched cutoff difference, and then estimate this coefficient. No such dispersion theorem, prime-pair asymptotic, positivity result, or breach of sieve parity is claimed here.

Keywords: primitive Möbius tail; canonical normalized determinant; support cardinality; translated determinant window; residue cell; Schur bound; dispersion gate.

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1 Introduction

The fixed-shift prime-pair problem separates local congruence compatibility from global signed cancellation. Sieve methods can encode the local constraints with great precision, but a two-target lower bound encounters the parity barrier long before those local constraints become the whole problem. The present paper does not claim to bypass that barrier [3]. It continues a narrower program: make the unresolved Möbius-tail correlation successively more rigid, prove power saving on every sector accessible to exact incidence geometry, and state the remaining signed input without disguising it as a consequence of the absolute estimates.

TPC-18 reduced a failed slice to a primitive off-diagonal two-row correlation with rows $m = \ell d$, orbit length J , row scale Q , and nonzero affine determinant $h(m - n)$ [8]. TPC-19 and TPC-20 gave exact determinant-frequency and matched-channel normal forms, while also identifying coefficient and zero-frequency gates that the formal transforms do not estimate [9, 10]. TPC-28 closed the selected low-conductor packet before centering; TPC-29 removed large reflected-content and a content-rich sparse wedge; and TPC-30 used the full gcd of the two targets to prune the large-target-content sectors of the three retained raw channels [12–14].

The last of these steps leaves

$$\mathcal{G}_{m,n}(j) := (mj + h, nj + h) \leq X^\eta, \tag{1}$$

with the fully coprime target sector as the sharp model. The proposed next decomposition was by translated determinant windows and determinant residue classes. There is, however, a definitional issue that must be settled before such a decomposition is legitimate.

If $r \mid mj + h$ and $s \mid nj + h$ are selected divisor variables, then $g = (r, s)$ depends on that selection. The quotient

$$\frac{m - n}{g}$$

is useful inside the expanded divisor channel, but it is not a label of the unexpanded raw kernel: the same row–orbit triple has many factorizations and therefore potentially many values of g . A cell decomposition made with this quotient cannot be inserted before the divisor sum without recording its multiplicity.

The factorization-independent replacement is

$$c = \mathcal{G}_{m,n}(j), \quad \Delta_{m,n}^\#(j) = \frac{m-n}{c}. \quad (2)$$

On the primitive orbit, TPC-30's Euclidean identity gives $c \mid m-n$, so $\Delta^\#$ is integral. More is true. Dividing the two targets by their full gcd produces coprime integers

$$U = \frac{mj+h}{c}, \quad V = \frac{nj+h}{c},$$

and direct subtraction gives

$$\boxed{mV - nU = h\Delta^\#}. \quad (3)$$

Thus $h\Delta^\#$ is the exact determinant of the canonically reduced target pair. This is the invariant used throughout the paper.

1.1 The new cellwise estimate

Fix an exact content c and a set Ω_c of allowed normalized determinants. Since the physical row integers are distinct, for a fixed left row m and a fixed $\delta \in \Omega_c$ there is at most one right row,

$$n = m - c\delta.$$

The same statement holds with left and right reversed. Hence the corresponding bipartite row graph has both degrees at most the number of allowed δ values in the physical difference range. The Schur test and the two physical row ℓ^2 bounds turn this degree bound into a weighted row estimate. Exact content c also confines the orbit variable to one residue class modulo c , giving the factor $1 + J/c$.

This proves the master estimate

$$\left| \mathcal{S}(c \leq Z, \Delta^\# \in \Omega_c) \right| \ll X^\varepsilon Q \log(2L) \sum_{c \leq Z} \left(1 + \frac{J}{c} \right) \mathcal{H}_c(\Omega_c), \quad (4)$$

where $\mathcal{H}_c$ is the number of allowed normalized determinants in the physical range. The estimate is before divisor-variable expansion and therefore applies to each of the two mixed row channels and the both-ultra row channel.

Two consequences are especially transparent. For arbitrary translated intervals I_c of length at most W ,

$$\left| \mathcal{S}(c \leq Z, \Delta^\# \in I_c) \right| \ll X^\varepsilon Q(W+1)(Z + J \log(2Z)). \quad (5)$$

For one residue class $a_c \pmod{q}$ at each content,

$$\left| \mathcal{S}(c \leq Z, \Delta^\# \equiv a_c \pmod{q}) \right| \ll X^\varepsilon X Q \left(\frac{Z}{X} + \frac{1}{Q} + \frac{1}{q} \right). \quad (6)$$

The first estimate is translation-invariant: its interval may be centered at $|\Delta^\#| \asymp Q/c$, far from the deleted near-row sector.

1.2 Removing the small-content restriction

The cell estimate and TPC-30 are complementary. Given a threshold C , the part with $c > C$ is bounded by the large-content theorem, whereas (4) treats $c \leq C$. For a single translated interval I of length W and a residue class $a \pmod{q}$ this gives

$$\left| \mathcal{S}(\Delta^\# \in I, \Delta^\# \equiv a \pmod{q}) \right|$$

$$\ll X^\varepsilon XQ \left[\frac{1}{J} + \frac{1}{C} + \left(1 + \frac{W}{q} \right) \left(\frac{C}{X} + \frac{1}{Q} \right) \right]. \quad (7)$$

This is an all-content theorem: no condition such as (1) remains in the event being estimated. Optimization of C gives a fixed power whenever $1 + W/q$ is polynomially smaller than Q .

For a residue class over the full normalized-determinant range, the dependence on c improves the count and yields

$$\left| \mathcal{S}(\Delta^\# \equiv a \pmod{q}) \right| \ll X^\varepsilon XQ \left(\frac{1}{J} + \frac{1}{C} + \frac{C}{X} + \frac{1}{Q} + \frac{1}{q} \right). \quad (8)$$

Every fixed exponent below $\min\{1 - \beta, \beta, \rho\}$ is therefore available when $q = X^{\rho+o(1)}$.

1.3 Why the theorem stops at one cell

The gain in (4) is cardinality gain. If a determinant range is partitioned into windows and every window is estimated absolutely, their cardinalities sum back to the size of the full range. Likewise, summing the absolute majorants over all $a \pmod{q}$ exactly restores all row pairs in the layer $c \mid m - n$. There is no hidden global saving.

For the signed physical amplitudes, the discrete Fourier transform in a is exact. Its $r = 0$ coefficient is the unweighted sum over all residue cells, namely the original exact-content contribution. This is an auxiliary residue-label transform, not the Poisson zero frequency in the orbit variable and not TPC-20's centered-kernel zero mode. Bounds for the nonzero frequencies or for equidistribution of the cells do not, by themselves, bound that mean. A candidate next route must construct a factorable representation of the prime-Möbius coefficient while retaining the fixed residue factors, pair mask, and matched channel structure; it cannot be obtained by merely repeating the present Schur estimate with a Fourier notation.

1.4 Scope

All estimates here are unconditional consequences of the inherited physical packet, TPC-30's proved large-content bound, and elementary finite-dimensional analysis. They do not use a Chowla-Elliott hypothesis, a prime-pair conjecture, or an unverified transfer of a Kloosterman or spectral large-sieve theorem. Fixed-form logarithmic two-point cancellation remains an important benchmark [7], but it does not provide the required uniformity in the polynomially growing row, content, cutoff, and mask families.

Section 2 states the physical interface. Section 3 proves the canonical reduction. Section 4 gives the row-graph theorem, and section 5 transfers it to the three raw channels. The all-content splice is in section 6; the high- β ledger is in section 7. The exact Fourier and reassembly boundary is isolated in section 8. The final sections compare the result with the preceding regions and record the finite certificate.

2 The inherited physical packet

Fix $h \in \mathbb{Z} \setminus \{0\}$ and put $H = \text{rad } |h|$. We use the primitive opened-row packet of TPC-25-TPC-30 [11, 12, 14]. The scales satisfy

$$QJ \asymp_{\mathcal{D}} X, \quad Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad 0 < \beta < 1. \quad (9)$$

Here $\mathcal{D}$ denotes fixed support, residue, smoothness, and margin data.

Rows are indexed by $\alpha = (\ell_\alpha, d_\alpha)$ and have integer labels

$$m_\alpha = \ell_\alpha d_\alpha \asymp_{\mathcal{D}} Q. \quad (10)$$

The opened packet has been separated so that m_α determines α uniquely. All row integers lie in an interval of length $O_{\mathcal{D}}(Q)$. Its two coefficient families obey

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}| \ll_{\mathcal{D}} Q, \quad (11)$$

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}|^2 \ll_{\mathcal{D}} Q \log(2L), \quad i = 1, 2. \quad (12)$$

The second estimate, rather than the first, is the source of the cellwise determinant gain.

Every contributing row and orbit point satisfies

$$(m_{\alpha}, H) = 1, \quad (j, H) = 1. \quad (13)$$

For

$$N_{\alpha}(j) = m_{\alpha}j + h, \quad (14)$$

the joint j -support lies in $O_{\mathcal{D}}(J)$ consecutive integers and

$$c_{\mathcal{D}}X \leq N_{\alpha}(j) \leq C_{\mathcal{D}}X. \quad (15)$$

In particular all targets are positive. Since H contains every prime divisor of h , (13) implies

$$(m_{\alpha}, h) = (j, h) = 1. \quad (16)$$

The row-pair mask satisfies

$$|\mathbf{m}(\alpha, \gamma)| \leq 1, \quad \mathbf{m}(\alpha, \gamma) = 0 \quad \text{if } m_{\alpha} = m_{\gamma}. \quad (17)$$

It is independent of the orbit and divisor variables. The inherited fixed-period factor $\Xi_{\alpha, \gamma}(j)$ and smooth weight $\mathcal{W}_{\alpha, \gamma}(j/J)$ are uniformly bounded; their common support has $O_{\mathcal{D}}(J)$ integer points.

2.1 The three retained raw channels

Let $\mathbf{A}_{m, T}(j)$ be the selected prefix residual and $C_m(j)$ the reflected ultra complement, in the exact notation of TPC-30. Their divisor coefficients and the elementary divisor bound give, uniformly on the physical support,

$$|\mathbf{A}_{m, T}(j)| + |C_m(j)| \ll_{\varepsilon, \mathcal{D}} X^{\varepsilon}. \quad (18)$$

The two mixed kernels and the both-ultra kernel are

$$\begin{aligned} \mathcal{K}_{\alpha, \gamma}^{(1)}(j) &= \mathbf{A}_{m_{\alpha}, T}(j) C_{m_{\gamma}}(j), \\ \mathcal{K}_{\alpha, \gamma}^{(2)}(j) &= C_{m_{\alpha}}(j) \mathbf{A}_{m_{\gamma}, T}(j), \\ \mathcal{K}_{\alpha, \gamma}^{(3)}(j) &= C_{m_{\alpha}}(j) C_{m_{\gamma}}(j). \end{aligned} \quad (19)$$

Thus

$$|\mathcal{K}_{\alpha, \gamma}^{(\nu)}(j)| \ll_{\varepsilon, \mathcal{D}} X^{\varepsilon}, \quad 1 \leq \nu \leq 3. \quad (20)$$

For an event $\mathcal{E}(\alpha, \gamma, j)$, define

$$\begin{aligned} \mathcal{S}_{\nu}(\mathcal{E}) &:= \sum_{\alpha, \gamma} \gamma_{\alpha}^{(1)} \gamma_{\gamma}^{(2)} \mathbf{m}(\alpha, \gamma) \sum_j \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J) \\ &\quad \times \mathcal{K}_{\alpha, \gamma}^{(\nu)}(j) \mathbf{1}_{\mathcal{E}(\alpha, \gamma, j)}. \end{aligned} \quad (21)$$

The natural absolute scale supplied by (11) and the orbit length is

$$JQ^2 \asymp XQ. \quad (22)$$

The two drift legs in the exact cutoff identity already carry the independent source-row saving established in TPC-29 [13]. The present paper changes none of their estimates. Every theorem below applies separately to each kernel in (19), and hence to their sum, without opening either divisor sum.

3 Canonical target reduction

The normalized determinant used here is attached to a row-orbit triple, not to one factorization of its targets.

Definition 3.1 (Full content and canonical determinant). For two distinct primitive rows m, n and a primitive orbit point j , put

$$c = \mathcal{G}_{m,n}(j) := (mj + h, nj + h), \quad \Delta_{m,n}^\#(j) := \frac{m-n}{c}. \quad (23)$$

When no confusion is possible we write simply $\Delta^\#$.

Proposition 3.2 (Canonical coprime target reduction). *Under the primitive hypotheses*

$$m \neq n, \quad (mn, h) = 1, \quad (j, h) = 1, \quad (24)$$

let c and $\Delta^\#$ be as in (23), and define

$$U = \frac{mj + h}{c}, \quad V = \frac{nj + h}{c}. \quad (25)$$

Then

$$c \mid m - n, \quad (c, mn h) = 1, \quad (26)$$

$$(U, V) = 1, \quad (27)$$

$$\boxed{mV - nU = h\Delta^\# \neq 0.} \quad (28)$$

In particular $\Delta^\# \in \mathbb{Z} \setminus \{0\}$ and is uniquely determined by (m, n, j) .

Proof. The primitive target-gcd identity of TPC-30 gives

$$c = (mj + h, m - n) \mid m - n \quad (29)$$

and $(c, mn h) = 1$ [14]. The two quotients in (25) are integers, and dividing two integers by their full gcd proves (27). Finally,

$$\begin{aligned} mV - nU &= \frac{m(nj + h) - n(mj + h)}{c} \\ &= \frac{h(m - n)}{c} = h\Delta^\#. \end{aligned}$$

The last quantity is nonzero because $h \neq 0$ and the row mask is off-diagonal. □

Remark 3.3 (A determinant of a coprime pair). Equation (28) may be written

$$\det \begin{pmatrix} m & U \\ n & V \end{pmatrix} = h\Delta^\#.$$

The columns are not asserted to be primitive as vectors, but the target column (U, V) is a coprime pair. This normalization uses every common target factor exactly once.

3.1 Relation to selected reflected divisors

Suppose a divisor expansion selects

$$r \mid mj + h, \quad s \mid nj + h, \quad d = (r, s). \quad (30)$$

TPC-29 attaches to the resulting two quotient forms the determinant $h(m - n)/d$ [13]. The following relation separates that factorization-dependent quantity from the canonical invariant.

Proposition 3.4 (Selected determinants are integral dilates). *Under the hypotheses of [proposition 3.2](#) and [equation \(30\)](#), one has*

$$d \mid c \quad (31)$$

and

$$\boxed{\frac{h(m - n)}{d} = h \frac{c}{d} \Delta^\#}. \quad (32)$$

Thus every selected determinant is an integral dilate of the canonical determinant. If $c = 1$, then $d = 1$ for every selected pair and all selected determinants equal $h\Delta^\#$.

Proof. The integer d divides both selected divisors, hence both targets. It therefore divides their full gcd c . Substitute $m - n = c\Delta^\#$. $\square$

Remark 3.5 (Why selected content cannot label the raw kernel). Different divisor pairs (r, s) of the same two targets may have different values of $d = (r, s)$. Consequently $(m - n)/d$ is not a single-valued function of (m, n, j) . By contrast, c and $\Delta^\#$ are unique before either kernel in (19) is expanded. This is why the physical cell theorems below use $\Delta^\#$.

Remark 3.6 (The small-content multiplier). On a layer $c \leq C$, the dilation factor c/d in (32) is at most C . Nevertheless a congruence for $\Delta^\#$ modulo q cannot be transported bijectively to the selected determinant unless the gcd of c/d and q is tracked. No such invertibility is assumed in this paper.

The exact content also controls the orbit occupancy. We record the form needed below.

Lemma 3.7 (One orbit class for an exact content). *Let $\mathcal{I}$ be an interval containing $O_{\mathcal{Q}}(J)$ consecutive integers. For fixed primitive rows $m \neq n$ and $c \geq 1$,*

$$\#\{j \in \mathcal{I} : (j, h) = 1, \mathcal{G}_{m,n}(j) = c\} \ll_{\mathcal{Q}} 1 + \frac{J}{c}. \quad (33)$$

The set is empty unless $c \mid m - n$; when nonempty, all of its points lie in the single class

$$j \equiv -h\overline{m} \pmod{c}. \quad (34)$$

Proof. This is the exact-content occupancy lemma of TPC-30. Indeed, $c \mid mj + h$ and $(c, m) = 1$ select (34); one class modulo c occurs in the interval at most $O_{\mathcal{Q}}(1 + J/c)$ times. Requiring the gcd to equal c , rather than merely to be divisible by c , only removes points. $\square$

4 Normalized-determinant graphs and support complexity

Let $\mathcal{M}$ be the finite set of physical row integers. It is contained in an interval of length at most $C_{\mathcal{Q}}Q$, and each integer in $\mathcal{M}$ labels at most one row. For $c \geq 1$ and $\Omega \subset \mathbb{Z} \setminus \{0\}$, define the bipartite adjacency matrix

$$A_{c,\Omega}(\alpha, \gamma) = \mathbf{1}_{(m_\alpha - m_\gamma)/c \in \Omega}. \quad (35)$$

The indicator already forces $c \mid m_\alpha - m_\gamma$. The exclusion of zero is consistent with the off-diagonal mask.

Definition 4.1 (Physical determinant support count). Choose once and for all a constant $B_{\mathcal{D}}$ larger than the row-span constant. Put

$$\mathcal{H}_c(\Omega) := \# \left(\Omega \cap \left[-\frac{B_{\mathcal{D}}Q}{c}, \frac{B_{\mathcal{D}}Q}{c} \right] \right). \quad (36)$$

We call this cardinality the determinant support count, or support complexity, of Ω at content c . Its logarithm would be the zero-order (Hartley) support entropy; no Shannon or metric entropy is used in the argument. The estimates below are stated directly in terms of the count $\mathcal{H}_c$.

Lemma 4.2 (Two-sided degree bound). *Every row and every column of $A_{c,\Omega}$ contains at most $\mathcal{H}_c(\Omega)$ nonzero entries.*

Proof. Fix a left row integer m . For each allowed δ , the equation

$$m - n = c\delta$$

determines $n = m - c\delta$ uniquely. Since row integers determine row labels uniquely, there is at most one right label for this value. Only values in the physical difference range can occur, so the row degree is at most (36). Interchanging m and n gives the same column bound. $\square$

Lemma 4.3 (Bipartite Schur bound). *Let $A = (A_{uv})$ be a finite zero-one matrix with maximum row degree K_+ and maximum column degree K_- . Then, for arbitrary complex vectors x and y ,*

$$\left| \sum_{u,v} x_u y_v A_{uv} \right|^2 \leq K_+ K_- \|x\|_2^2 \|y\|_2^2. \quad (37)$$

Proof. For each u , Cauchy–Schwarz gives

$$\left| \sum_v A_{uv} y_v \right|^2 \leq K_+ \sum_v A_{uv} |y_v|^2.$$

Summing in u and using the column-degree bound yields

$$\|Ay\|_2^2 \leq K_+ K_- \|y\|_2^2.$$

Apply Cauchy–Schwarz to $\langle x, Ay \rangle$. $\square$

Combining the preceding two lemmas with the physical coefficient bound gives the row estimate used in every later section.

Proposition 4.4 (Weighted determinant-support bound). *For every $c \geq 1$ and $\Omega \subset \mathbb{Z} \setminus \{0\}$,*

$$\boxed{\sum_{\alpha, \gamma} |\gamma_{\alpha}^{(1)} \gamma_{\gamma}^{(2)}| A_{c,\Omega}(\alpha, \gamma) \ll_{\mathcal{D}} Q \log(2L) \mathcal{H}_c(\Omega).} \quad (38)$$

Proof. Apply lemma 4.3 to the nonnegative vectors $|\gamma^{(1)}|$ and $|\gamma^{(2)}|$, use lemma 4.2, and then use (12) for the two norms. $\square$

4.1 Three useful support-count calculations

The first calculation is independent of the location of a window.

Lemma 4.5 (Translated windows and residue cells). *Let $I \subset \mathbb{R}$ be any interval of length at most $W \geq 0$. Then*

$$\mathcal{H}_c(I \cap \mathbb{Z}) \leq W + 1, \quad (39)$$

$$\mathcal{H}_c(\{\delta \in I \cap \mathbb{Z} : \delta \equiv a \pmod{q}\}) \leq 1 + \frac{W}{q} \quad (40)$$

for every $q \geq 1$ and $a \pmod{q}$.

Proof. An interval of length W contains at most $W + 1$ integers. Integers in one class modulo q are spaced by q , so their number in the same interval is at most $1 + W/q$. Intersecting with the physical range can only reduce either count. $\square$

For a residue class without an imposed window, the physical range retains the content dependence.

Lemma 4.6 (Full-range residue support). *Uniformly for $q \geq 1$ and $a \pmod{q}$,*

$$\boxed{\mathcal{H}_c(\{\delta \in \mathbb{Z} : \delta \equiv a \pmod{q}\}) \ll_{\mathcal{D}} 1 + \frac{Q}{cq}.} \quad (41)$$

No coprimality condition between c and q is needed.

Proof. The physical normalized-determinant interval has length $O_{\mathcal{D}}(Q/c)$. One class modulo q occupies it at most $O_{\mathcal{D}}(1 + Q/(cq))$ times. $\square$

Remark 4.7 (The endpoint one cannot be removed). If q or c is larger than the relevant span, a nonempty residue cell may still contain one normalized determinant. The $+1$ terms in (40) and (41) are therefore genuine floors of the deterministic graph estimate.

Remark 4.8 (Macroscopic translation). Nothing in lemma 4.5 depends on the distance from I to zero. One may take $I \subset [\kappa Q/c, 2\kappa Q/c]$ for a fixed $\kappa > 0$ whenever that interval meets the physical difference range. The resulting estimate is not another deletion of near-diagonal rows.

5 Physical transfer on the small-content side

For a physical triple abbreviate

$$c_{\alpha,\gamma}(j) := \mathcal{G}_{m_\alpha, m_\gamma}(j), \quad \Delta_{\alpha,\gamma}^\#(j) := \frac{m_\alpha - m_\gamma}{c_{\alpha,\gamma}(j)}. \quad (42)$$

Both quantities are defined before opening any divisor sum.

Let $Z \geq 1$, and for each integer $1 \leq c \leq Z$ choose an arbitrary set $\Omega_c \subset \mathbb{Z} \setminus \{0\}$. Define the disjoint events

$$\mathcal{E}_c(\Omega_c) := \{c_{\alpha,\gamma}(j) = c, \Delta_{\alpha,\gamma}^\#(j) \in \Omega_c\} \quad (43)$$

and

$$\mathcal{E}_{\leq Z}(\Omega) := \bigcup_{1 \leq c \leq Z} \mathcal{E}_c(\Omega_c). \quad (44)$$

Theorem 5.1 (Small-content determinant-support transfer). *For every $1 \leq \nu \leq 3$,*

$$\boxed{|\mathcal{S}_\nu(\mathcal{E}_{\leq Z}(\Omega))| \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon Q \log(2L) \sum_{1 \leq c \leq Z} \left(1 + \frac{J}{c}\right) \mathcal{H}_c(\Omega_c).} \quad (45)$$

The same bound, with a changed constant, holds for the sum of the three raw channels.

Proof. Take absolute values and split the disjoint union by its exact content c . For a fixed surviving row pair, [lemma 3.7](#) leaves at most $O_{\mathcal{D}}(1+J/c)$ orbit points. The kernel, periodic factor, smooth weight, and mask have total pointwise size $O(X^\varepsilon)$ by [\(17\)](#) and [\(20\)](#). The remaining row pair must satisfy

$$\frac{m_\alpha - m_\gamma}{c} \in \Omega_c.$$

Apply [proposition 4.4](#) and sum over c . There are only three kernels. $\square$

Remark 5.2 (No divisor multiplicity). The proof bounds each complete raw kernel through [\(20\)](#). It neither chooses a reflected divisor pair nor sums over selected gcd values. Exact target content is single-valued, so [\(45\)](#) contains no unrecorded factorization multiplicity.

5.1 Translated determinant windows

Let $I_c \subset \mathbb{R}$ be arbitrary intervals, each of length at most W . Apply [\(39\)](#) and the elementary bounds

$$\sum_{c \leq Z} 1 \leq Z, \quad \sum_{c \leq Z} \frac{1}{c} \ll \log(2Z). \quad (46)$$

Corollary 5.3 (Uniform translated-window pruning). *For every $1 \leq \nu \leq 3$,*

$$|\mathcal{S}_\nu(c_{\alpha,\gamma}(j) \leq Z, \Delta_{\alpha,\gamma}^\#(j) \in I_c)| \ll_{\varepsilon,h,\mathcal{D}} X^\varepsilon Q \log(2L)(W+1)(Z+J \log(2Z)). \quad (47)$$

Here the interval is allowed to depend on the exact content c .

Proof. Use [theorem 5.1](#) with $\Omega_c = I_c \cap \mathbb{Z}$ and apply [equations \(39\)](#) and [\(46\)](#). $\square$

In power form, suppose for fixed $z, \omega \geq 0$ that

$$Z \leq X^{z+o(1)}, \quad W+1 \leq X^{\omega+o(1)}. \quad (48)$$

Relative to $XQ \asymp JQ^2$, the two terms in [\(47\)](#) have ratios

$$X^{\omega+z-1+o(1)}, \quad X^{\omega-\beta+o(1)}. \quad (49)$$

Corollary 5.4 (Small-content window saving). *If*

$$\omega < \min\{1-z, \beta\}, \quad (50)$$

then every fixed

$$\sigma < \min\{1-z, \beta\} - \omega \quad (51)$$

is a power-saving exponent relative to XQ for the sum of the three raw channels in [\(47\)](#).

Proof. Use [\(9\)](#) in [\(49\)](#); the strict inequalities absorb the logarithms and all scale $o(1)$ terms. $\square$

In particular, the fully coprime layer has $c = 1$ and no content sum.

Corollary 5.5 (Fully coprime target window). *For any translated interval I of length at most W ,*

$$|\mathcal{S}_\nu(c_{\alpha,\gamma}(j) = 1, \Delta_{\alpha,\gamma}^\#(j) \in I)| \ll_{\varepsilon,h,\mathcal{D}} X^\varepsilon JQ \log(2L)(W+1). \quad (52)$$

Thus, if $W = X^{\omega+o(1)}$ and $\omega < \beta$, every fixed saving below $\beta - \omega$ is available on this layer.

Proof. Set $Z = 1$ in [theorem 5.1](#) and use $1+J \ll J$ on the asymptotic physical range. Since $JQ \asymp X$, the ratio to XQ is $X^{\omega-\beta+o(1)}$. $\square$

5.2 Individual residue cells

For each $c \leq Z$, choose a residue $a_c \pmod{q}$. By [lemma 4.6](#),

$$\begin{aligned} & \sum_{c \leq Z} \left(1 + \frac{J}{c}\right) \left(1 + \frac{Q}{cq}\right) \\ & \ll Z + J \log(2Z) + \frac{Q}{q} \log(2Z) + \frac{JQ}{q}. \end{aligned} \quad (53)$$

Corollary 5.6 (Small-content residue-cell pruning). *For every $1 \leq \nu \leq 3$,*

$$\begin{aligned} & |\mathcal{S}_\nu(c_{\alpha,\gamma}(j) \leq Z, \Delta_{\alpha,\gamma}^\#(j) \equiv a_c \pmod{q})| \\ & \ll_{\varepsilon,h,\mathcal{D}} X^\varepsilon \log(2L) \left(QZ + JQ \log(2Z) + \frac{Q^2}{q} \log(2Z) + \frac{JQ^2}{q} \right) \end{aligned} \quad (54)$$

$$\ll_{\varepsilon,h,\mathcal{D}} X^\varepsilon XQ \left(\frac{Z}{X} + \frac{1}{Q} + \frac{1}{q} \right). \quad (55)$$

The residue a_c may depend on c .

Proof. Use [theorem 5.1](#) and (53). In the second form, enlarge ε to absorb logarithms, use $JQ \asymp X$, and note that $1/(Jq) \leq 1/q$. $\square$

If $Z \leq X^{z+o(1)}$ with $z < 1$ and $q = X^{\rho+o(1)}$, every fixed

$$\sigma < \min\{1 - z, \beta, \rho\} \quad (56)$$

is available. For the fully coprime target layer this specializes to

$$|\mathcal{S}_\nu(c = 1, \Delta^\# \equiv a \pmod{q})| \ll X^\varepsilon XQ \left(\frac{1}{Q} + \frac{1}{q} \right). \quad (57)$$

5.3 A window inside one residue class

Combining (40) with the master theorem gives the most localized form.

Corollary 5.7 (Small-content window–residue cell). *Let I_c have length at most W , and choose $a_c \pmod{q}$. Then*

$$\begin{aligned} & |\mathcal{S}_\nu(c_{\alpha,\gamma}(j) \leq Z, \Delta_{\alpha,\gamma}^\#(j) \in I_c, \Delta_{\alpha,\gamma}^\#(j) \equiv a_c \pmod{q})| \\ & \ll_{\varepsilon,h,\mathcal{D}} X^\varepsilon Q \log(2L) \left(1 + \frac{W}{q} \right) (Z + J \log(2Z)). \end{aligned} \quad (58)$$

The bound depends only on the number of possible determinant values in the intersection, not on the location of the window or on (c, q) .

6 Splicing large content to low-complexity cells

The preceding section assumes $c \leq Z$. TPC-30 supplies the complementary estimate

$$|\mathcal{S}_\nu(c > C)| \ll_{\varepsilon,h,\mathcal{D}} X^\varepsilon \left(Q^2 + \frac{XQ}{C} \right) \quad (59)$$

for every $C \geq 1$ [14]. Any further determinant restriction only reduces the absolute majorant in that theorem. We now splice (59) to the small-content cell estimate.

6.1 A translated window in one residue class

Let $I \subset \mathbb{R}$ be any interval of length at most W , and fix one class $a \pmod{q}$. Define the factorization-independent all-content event

$$\mathcal{E}(I; a, q) := \{\Delta_{\alpha, \gamma}^{\#}(j) \in I, \Delta_{\alpha, \gamma}^{\#}(j) \equiv a \pmod{q}\}. \quad (60)$$

Theorem 6.1 (All-content determinant-cell pruning). *For every $1 \leq C \ll_{\mathscr{D}} Q$ and $1 \leq \nu \leq 3$,*

$$|\mathcal{S}_{\nu}(\mathcal{E}(I; a, q))| \ll_{\varepsilon, h, \mathscr{D}} X^{\varepsilon} \left\{ Q^2 + \frac{XQ}{C} + Q \log(2L) \left(1 + \frac{W}{q}\right) (C + J \log(2C)) \right\} \quad (61)$$

$$\ll_{\varepsilon, h, \mathscr{D}} X^{\varepsilon} XQ \left[\frac{1}{J} + \frac{1}{C} + \left(1 + \frac{W}{q}\right) \left(\frac{C}{X} + \frac{1}{Q}\right) \right]. \quad (62)$$

The same estimate holds for the sum of the three raw channels.

Proof. Split the event according to $c > C$ and $c \leq C$. The first part is a subevent of (59). For the second, apply corollary 5.7 with the same interval and residue class for every $c \leq C$. This proves the expanded form. Use $JQ \asymp X$ and absorb logarithms into X^{ε} to obtain (62). $\square$

The threshold C is an optimization variable, not a condition in the event. We give the exponent consequence with fixed margins.

Corollary 6.2 (Optimized all-content cell saving). *Suppose*

$$\frac{W}{q} + 1 \leq X^{s+o(1)} \quad (63)$$

for a fixed $s \geq 0$. If $s < \beta$, then every fixed

$$\boxed{\sigma < \min\{1 - \beta, \beta - s\}} \quad (64)$$

is a power-saving exponent relative to XQ in (62).

Proof. Fix a σ satisfying (64). Choose κ strictly between

$$\sigma < \kappa < \min\{\beta, 1 - s - \sigma\}. \quad (65)$$

This interval is nonempty: $\sigma < \beta$, and adding $\sigma < 1 - \beta$ to $\sigma < \beta - s$ gives

$$2\sigma < 1 - s. \quad (66)$$

Take $C = \lfloor X^{\kappa} \rfloor$. The four ratios in (62) then save respectively $1 - \beta$, κ , $1 - \kappa - s$, and $\beta - s$, each strictly more than σ , up to $o(1)$. The strict margins absorb the scale and logarithmic losses. $\square$

Remark 6.3 (Endpoint convention). Equation (64) is a relaxed margin: every fixed exponent below the displayed endpoint is proved. The endpoint itself is not asserted, because the physical scales contain $o(1)$ factors and the estimates contain logarithms.

For an interval alone, take $q = 1$. If $W = X^{\omega+o(1)}$ and $\omega < \beta$, the relaxed all-content saving is

$$\min\{1 - \beta, \beta - \omega\}. \quad (67)$$

The interval may be macroscopically separated from zero.

Such separation itself gives a further content collapse.

Proposition 6.4 (Macro-centered content collapse). *Fix $\lambda > 0$, and suppose the translated interval I lies in*

$$\{\delta \in \mathbb{R} : |\delta| \geq \lambda Q\}. \quad (68)$$

Then every physical triple with $\Delta^\# \in I$ has

$$c_{\alpha,\gamma}(j) \ll_{\lambda,\mathcal{D}} 1. \quad (69)$$

Consequently, for an interval of length at most W and one class $a \pmod{q}$,

$$\boxed{|\mathcal{S}_\nu(\mathcal{E}(I; a, q))| \ll_{\varepsilon,h,\lambda,\mathcal{D}} X^\varepsilon JQ \log(2L) \left(1 + \frac{W}{q}\right)}. \quad (70)$$

If $1 + W/q \leq X^{s+o(1)}$ with $s < \beta$, every fixed saving below $\beta - s$ is available relative to XQ .

Proof. The physical row span gives $|m_\alpha - m_\gamma| \leq B_{\mathcal{D}}Q$. Since $m_\alpha - m_\gamma = c_{\alpha,\gamma}(j)\Delta^\#$, (68) implies $c_{\alpha,\gamma}(j) \leq B_{\mathcal{D}}/\lambda$. Apply [corollary 5.7](#) with this fixed content ceiling. The resulting content sum is $O_{\lambda,\mathcal{D}}(J)$, proving (70). Divide by $XQ \asymp JQ^2$ for the exponent statement. $\square$

6.2 One residue class over the full range

Let

$$\mathcal{E}(a, q) := \{\Delta_{\alpha,\gamma}^\#(j) \equiv a \pmod{q}\}. \quad (71)$$

Using the content-dependent range in (41) is sharper than enclosing every content in one interval of length $O(Q)$.

Theorem 6.5 (All-content residue-cell pruning). *For every $1 \leq C \ll_{\mathcal{D}} Q$ and $1 \leq \nu \leq 3$,*

$$|\mathcal{S}_\nu(\mathcal{E}(a, q))| \ll_{\varepsilon,h,\mathcal{D}} X^\varepsilon \left(Q^2 + \frac{XQ}{C} + QC + JQ \log(2C) + \frac{Q^2}{q} \log(2C) + \frac{JQ^2}{q} \right) \quad (72)$$

$$\ll_{\varepsilon,h,\mathcal{D}} X^\varepsilon XQ \left(\frac{1}{J} + \frac{1}{C} + \frac{C}{X} + \frac{1}{Q} + \frac{1}{q} \right). \quad (73)$$

Proof. Use (59) for $c > C$ and [corollary 5.6](#) with $Z = C$ for the complementary part. The simplified form follows as in [theorem 6.1](#). $\square$

Corollary 6.6 (Optimized residue-cell saving). *If $q = X^{\rho+o(1)}$ with $\rho \geq 0$, then every fixed*

$$\boxed{\sigma < \min\{1 - \beta, \beta, \rho\}} \quad (74)$$

is a power-saving exponent relative to XQ , provided the minimum is positive.

Proof. Fix a σ satisfying (74). Since $\sigma < \min\{\beta, 1 - \beta\}$, choose

$$\sigma < \kappa < \min\{\beta, 1 - \sigma\}. \quad (75)$$

Take $C = \lfloor X^\kappa \rfloor$. In (73), the five displayed ratios save $1 - \beta$, κ , $1 - \kappa$, β , and ρ , respectively; all exceed σ . The strict margins absorb the endpoint losses. $\square$

Remark 6.7 (What the splice adds). TPC-30 alone estimates the event $c > C$ without seeing determinant support complexity. The small-content theorem alone estimates a determinant cell only after imposing $c \leq C$. Their splice proves a cellwise statement for the entire retained raw packet. It does not estimate the union of all cells with a power saving.

7 The high- β determinant-cell ledger

We insert the all-content theorems into the selected high- β packet closed in TPC-28. Its exact exponents are

$$\beta = \frac{267}{400}, \quad j = 1 - \beta = \frac{133}{400}. \quad (76)$$

Thus

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}. \quad (77)$$

Every conclusion in this section is an upper bound for a support-compatible cell. It does not assert that the cell is nonempty or has positive asymptotic mass.

7.1 A translated orbit-scale window

Let I_X be any interval satisfying

$$|I_X| = X^{133/400+o(1)} = JX^{o(1)}. \quad (78)$$

The interval is initially allowed any center. For the fully coprime target layer, [corollary 5.5](#) gives the relaxed margin

$$\beta - j = \frac{267}{400} - \frac{133}{400} = \frac{134}{400} = \frac{67}{200}. \quad (79)$$

For all target contents, use [\(67\)](#). The large-content side limits the splice to

$$\min \left\{ \frac{133}{400}, \frac{134}{400} \right\} = \left\lfloor \frac{133}{400} \right\rfloor. \quad (80)$$

Corollary 7.1 (All-content orbit-scale translated window). *For every fixed $\sigma < 133/400$, the sum of the three physical raw channels restricted by*

$$\Delta_{\alpha,\gamma}^{\#}(j) \in I_X \quad (81)$$

satisfies

$$\sum_{\nu=1}^3 |\mathcal{S}_{\nu}(\Delta^{\#} \in I_X)| \ll_{\sigma,h,\mathcal{D}} XQX^{-\sigma}. \quad (82)$$

Proof. In [corollary 6.2](#), take $q = 1$ and $s = 133/400$. Equation [\(80\)](#) is the resulting endpoint. The $o(1)$ in [\(78\)](#) is absorbed by the strict choice of σ . $\square$

If the same window is genuinely macro-centered, the all-content result improves because large content cannot occur.

Corollary 7.2 (Macro-centered orbit-scale window). *Assume in addition that, for a fixed $\lambda > 0$,*

$$I_X \subset \{\delta : |\delta| \geq \lambda Q\}. \quad (83)$$

Then every fixed $\sigma < 67/200$ satisfies

$$\sum_{\nu=1}^3 |\mathcal{S}_{\nu}(\Delta^{\#} \in I_X)| \ll_{\sigma,h,\lambda,\mathcal{D}} XQX^{-\sigma}. \quad (84)$$

Proof. Use [proposition 6.4](#) with $q = 1$ and support exponent $s = 133/400$. The available margin is $\beta - s = 134/400 = 67/200$. $\square$

7.2 A wider macroscopic window

Take instead

$$|I_X| = X^{1/2+o(1)} \quad (85)$$

and again allow the center to be $\asymp Q$. Since

$$\beta - \frac{1}{2} = \frac{267}{400} - \frac{200}{400} = \boxed{\frac{67}{400}}, \quad (86)$$

the cell remains power-small despite containing many more than J normalized determinant values.

Corollary 7.3 (All-content half-scale translated window). *For every fixed $\sigma < 67/400$,*

$$\boxed{\sum_{\nu=1}^3 |\mathcal{S}_\nu(\Delta^\# \in I_X)| \ll_{\sigma, h, \mathcal{D}} X Q X^{-\sigma}} \quad (87)$$

whenever (85) holds.

Proof. Use (67) with window exponent $1/2$:

$$\min \left\{ \frac{133}{400}, \frac{267}{400} - \frac{1}{2} \right\} = \frac{67}{400}.$$

□

7.3 One orbit-scale residue class

Let q_X be any integer sequence with

$$q_X = X^{133/400+o(1)}. \quad (88)$$

No primality or coprimality condition on q_X is needed for the canonical determinant cell. For every choice of $a_X \pmod{q_X}$, [corollary 6.6](#) gives

$$\min \left\{ \frac{133}{400}, \frac{267}{400}, \frac{133}{400} \right\} = \boxed{\frac{133}{400}}. \quad (89)$$

Corollary 7.4 (All-content orbit-scale residue cell). *For every fixed $\sigma < 133/400$,*

$$\boxed{\sum_{\nu=1}^3 |\mathcal{S}_\nu(\Delta^\# \equiv a_X \pmod{q_X})| \ll_{\sigma, h, \mathcal{D}} X Q X^{-\sigma}}. \quad (90)$$

The intersection of an interval of length $\asymp J$ with one class modulo $q_X \asymp J$ contains $O(1)$ integers. The general all-content splice has margin $133/400$. On the fully coprime layer alone the margin is $\beta = 267/400$; and if the interval is macro-centered as in (83), [proposition 6.4](#) shows that the same $267/400$ margin holds for all contents. Without that location hypothesis, the stronger margin must not be quoted for the all-content cell.

7.4 Separation from earlier deletions

The translated intervals above may be chosen inside

$$\kappa Q \leq |\Delta^\#| \leq 2\kappa Q \quad (91)$$

for fixed $\kappa > 0$, whenever the physical row support meets that annulus. Then [proposition 6.4](#) gives $c \ll_{\kappa, \mathcal{D}} 1$ and $|m_\alpha - m_\gamma| \asymp Q$, so this lies outside a near-row deletion. On the fully coprime sublayer, every selected reflected gcd equals one by [proposition 3.4](#); the large-full-content theorem of TPC-30 and the large-reflected-content part of TPC-29 do not estimate that subcell. Some expanded divisor subfamilies may overlap a different TPC-29 sparse region, but the present bound is uniform for the complete unexpanded raw kernel and does not assume that sparse condition.

Table 1: Margins supplied by the high- β determinant-cell bounds.

Cell	Available relaxed margin	Source
Fully coprime J -window	67/200	row support count J/Q
General all-content J -window	133/400	large-content splice
Macro-centered all-content J -window	67/200	bounded-content collapse
All-content $X^{1/2}$ -window	67/400	window cardinality
All-content class modulo J	133/400	modulus/splice
General all-content intersection: $ I \asymp q \asymp J$	133/400	large-content splice
Macro-centered all-content intersection	267/400	bounded-content one-cell floor

Each table entry is the relaxed margin supplied by the stated bound, not an optimality claim. Every strictly smaller fixed exponent is proved; no endpoint equality is claimed.

8 Cell energy, exceptional frequencies, and the auxiliary zero coefficient

The cell theorems imply a useful mean-square statement, but they also show exactly why that statement does not estimate the full residual. Fix $C \geq 1$, $q \geq 1$, one raw channel ν , and define the signed small-content cell amplitudes

$$S_a := \mathcal{S}_\nu(c_{\alpha, \gamma}(j) \leq C, \Delta_{\alpha, \gamma}^\#(j) \equiv a \pmod{q}), \quad a \pmod{q}. \quad (92)$$

Let A_a be the corresponding absolute majorant obtained from (21) by replacing every coefficient, kernel, mask, and weight by its absolute value. Then

$$|S_a| \leq A_a. \quad (93)$$

Remark 8.1 (Zero-mode nomenclature). The transform below is a finite Fourier transform in the auxiliary residue label $a \pmod{q}$. Its $r = 0$ coefficient is not the Poisson zero frequency in the orbit variable j , and it is not the centered divisor-kernel zero mode of TPC-20. We use “residue-cell zero coefficient” when this distinction matters.

Because residue classes partition the exact-content triples,

$$\sum_{a \pmod{q}} A_a \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon J Q^2. \quad (94)$$

On the other hand, the proof of [corollary 5.6](#) is an absolute proof and gives

$$\max_{a(q)} A_a \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon JQ^2 \left(\frac{C}{X} + \frac{1}{Q} + \frac{1}{q} \right). \quad (95)$$

Proposition 8.2 (Residue-cell energy). *The signed cells in (92) satisfy*

$$\boxed{\sum_{a(q)} |S_a|^2 \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon (JQ^2)^2 \left(\frac{C}{X} + \frac{1}{Q} + \frac{1}{q} \right)}. \quad (96)$$

If $C \ll J$, this simplifies to

$$\sum_{a(q)} |S_a|^2 \ll X^\varepsilon (JQ^2)^2 \left(\frac{1}{Q} + \frac{1}{q} \right). \quad (97)$$

Proof. By (93),

$$\sum_a |S_a|^2 \leq \sum_a A_a^2 \leq (\max_a A_a) \sum_a A_a.$$

Apply (94) and (95), each with $\varepsilon/2$, and relabel the resulting soft loss as X^ε . If $C \ll J$, then $C/X \ll 1/Q$ because $JQ \asymp X$. $\square$

Define the unnormalized discrete Fourier transform

$$\widehat{S}(r) := \sum_{a(q)} S_a e_q(-ra), \quad e_q(x) = \exp(2\pi i x/q). \quad (98)$$

Since each triple lies in the class $a \equiv \Delta^\# \pmod{q}$, this transform also has the exact physical form

$$\widehat{S}(r) = \mathcal{S}_\nu(c \leq C; e_q(-r\Delta^\#)), \quad (99)$$

where the notation on the right means that the phase is inserted into the raw summand.

Corollary 8.3 (Mean-square frequency compression). *If $C \ll J$, then*

$$\boxed{\frac{1}{q} \sum_{r(q)} |\widehat{S}(r)|^2 \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon (JQ^2)^2 \left(\frac{1}{Q} + \frac{1}{q} \right)}. \quad (100)$$

In particular, if $q \leq Q$ and $\sigma > 0$, then

$$\#\{r \pmod{q} : |\widehat{S}(r)| > JQ^2 X^{-\sigma}\} \ll_{\varepsilon, h, \mathcal{D}} X^{2\sigma+\varepsilon}. \quad (101)$$

Proof. Parseval for (98) gives

$$\frac{1}{q} \sum_r |\widehat{S}(r)|^2 = \sum_a |S_a|^2.$$

Use (97). If $q \leq Q$, then

$$\sum_r |\widehat{S}(r)|^2 \ll X^\varepsilon (JQ^2)^2.$$

Chebyshev's inequality gives (101). $\square$

More precisely, if $q = X^{\rho+o(1)} \leq Q$ and $0 < \sigma < \rho/2$, then one may choose the soft-loss exponent with $0 < \varepsilon < \rho - 2\sigma$. The exceptional set in (101) is then power-smaller than the full set of q frequencies. This few-large-frequency bound is not, however, a bound for the coefficient needed by the original unweighted reassembly.

Proposition 8.4 (The exact residue-cell zero coefficient). *With the preceding notation,*

$$\boxed{\hat{S}(0) = \sum_{a(q)} S_a = \mathcal{S}_\nu(c_{\alpha,\gamma}(j) \leq C).} \quad (102)$$

Moreover, inversion gives

$$S_a - \frac{\hat{S}(0)}{q} = \frac{1}{q} \sum_{r=1}^{q-1} \hat{S}(r) e_q(ra), \quad (103)$$

and hence

$$\sum_{a(q)} \left| S_a - \frac{\hat{S}(0)}{q} \right|^2 = \frac{1}{q} \sum_{r=1}^{q-1} |\hat{S}(r)|^2. \quad (104)$$

Control of the nonzero-frequency side alone places no upper bound on $\hat{S}(0)$.

Proof. The first equality is the $r = 0$ case of (98); the second follows because the residue cells partition the event $c \leq C$. Fourier inversion proves (103), and Parseval proves (104). Finally, a constant vector $S_a = M/q$ has every nonzero Fourier coefficient equal to zero but has $\hat{S}(0) = M$. Thus no inequality involving only the nonzero frequencies can control the residue-cell zero coefficient without an additional input. $\square$

8.1 Absolute reassembly is exactly lossless

For the nonnegative majorants there is an even simpler obstruction:

$$\boxed{\sum_{a(q)} A_a = A_{\text{all}},} \quad (105)$$

where A_{all} is the absolute mass of the whole small-content event. Likewise, if a normalized-determinant interval is partitioned into disjoint integer windows, the sum of their determinant cardinalities equals the cardinality of the original interval. Consequently the local gains q^{-1} and W/Q disappear when all cells are recombined by the triangle inequality. This is an exact partition identity, not an assertion that the signed physical sum is large.

The next analytic obligation is therefore sharper than “prove that most nonzero determinant frequencies are small.” One candidate is to construct a factorable representation of the prime–Möbius coefficient while retaining the fixed residue factors, pair mask, and matched channel structure, and then control the residue-cell zero coefficient or cancel it across the mixed and both-ultra channels before separation. The exact channel recombination of TPC-20 and the coefficient gates isolated in TPC-19 provide relevant normal forms [9, 10]; the present paper does not supply the required signed estimate.

9 Comparison, sharpness, and the next gate

9.1 What is inherited and what is new

The Euclidean identity

$$(mj + h, nj + h) \mid m - n$$

and the single exact-content orbit class are inherited from TPC-30, with their earlier geometric origin in TPC-18 [8, 14]. TPC-19 already constructed determinant Fourier normal forms for an earlier centered model, and TPC-29 already computed the affine determinant associated with a selected reflected divisor pair [9, 13]. We claim none of those identities as new.

The additions here are:

- (i) the factorization-independent reduced pair (U, V) and its canonical determinant $mV - nU = h\Delta^\#$;
- (ii) the two-sided degree and Schur bounds for arbitrary canonical determinant spectra Ω_c ;
- (iii) their transfer to all three unexpanded ultra-long raw channels;
- (iv) translated-window, residue-cell, and window-residue bounds on the complete small-content family;
- (v) the splice with TPC-30, producing all-content single-cell estimates with optimized fixed-power margins;
- (vi) the residue-cell energy and exceptional-frequency compression;
- (vii) the exact identification of the small-content residue-cell reassembly as the $r = 0$ coefficient of an auxiliary residue-label DFT.

The word “determinant” here refers to the elementary 2×2 identity (28). The method is not the determinant method for counting rational or integral points on algebraic varieties [4]. Its counting input is only the injectivity of $n = m - c\delta$ and a bipartite Schur estimate.

9.2 Why the regions are genuinely different

TPC-18 deleted a neighborhood of the row diagonal. The present windows are translation-invariant and may be centered where $|m - n| \asymp Q$. TPC-29 works after selecting divisor variables and uses their gcd and lcm. The present cell is defined before divisor selection and remains meaningful for the whole raw kernel. TPC-30 deletes large full target content; the fully coprime layer $c = 1$ is the sharp model for the new theorem.

The all-content splice does overlap TPC-30 on $c > C$ by design. Its new input is the low-complexity determinant estimate on $c \leq C$; neither half alone yields theorem 6.1. On expanded divisor subfamilies there may also be overlap with one of TPC-29’s sparse wedges. The theorem does not assert disjointness from every such subfamily. It asserts a uniform raw-kernel bound requiring none of their selected-divisor hypotheses.

9.3 Sharp deterministic floors

The Q^{-1} term in a residue cell comes from one possible row match per left row. It cannot be removed from a theorem assuming only row uniqueness and ℓ^2 control. For example, take consecutive abstract row sets and the graph $m - n = 1$. Its degree is one, and constant nonnegative row weights attain the order of the Schur bound. In the physical target geometry, the same difference has exact content one at every primitive orbit point, because the target gcd divides $|m - n| = 1$.

Likewise, a translated window containing K integer determinants can have degree K on an interval of consecutive rows. The cardinality factor in (38) is therefore optimal under the abstract physical interface up to support edges and logarithmic normalization. This is not an optimality theorem for the signed opened-row coefficients; their arithmetic provenance has not yet been used.

9.4 The refined route after TPC-31

The Fourier calculation in section 8 refines the phrase “signed dispersion across determinant cells.” Even the few-large-frequency estimate does not control the original sum, because that sum is exactly the $r = 0$ coefficient $\hat{S}(0)$ of the auxiliary residue-label transform. The next stage must make this coefficient non-generic by using structure suppressed in the Schur majorant.

The resulting staged route is:

- TPC-32: construct a factorable representation of the prime–Möbius coefficient, incorporating $\mu(d)(\log \ell)\omega_D(d)\psi_L(\ell/L)$ inside the canonical determinant coefficient while retaining the fixed residue factors, the bounded pair mask, and the remaining physical weights; derive a matched TT^* or spectral dispersion form without separating the two mixed channels from the both-ultra channel;
- TPC-33: prove cancellation of the auxiliary zero coefficient or a deconcentration inequality strong enough to retain a fixed part of the cellwise margin;
- TPC-34: reassemble the exact cutoff difference and propagate the estimate uniformly over the physical parameter box;
- TPC-35: return to the complete TPC-18 residual and endpoint packets only after the preceding uniformity is proved.

The exact matched-channel identity of TPC-20 is a natural algebraic interface for the first step [10]. Classical large-sieve technology and modern Kloosterman-fraction or exceptional-spectrum estimates provide benchmarks [1, 2, 5, 6]. They cannot be inserted merely by analogy: the actual coefficient, parameter range, mask, and auxiliary zero coefficient must first be reduced to the precise hypotheses of one stated theorem. No such transfer is used in the present estimates.

Object	Status
Primitive target-gcd identity and exact-content orbit class	Inherited from TPC-30.
Canonical reduced targets and determinant $h\Delta^\#$	Factorization-independent canonical packaging proved here.
Arbitrary translated canonical determinant window	Explicit cellwise bound; power saving under the stated support-count condition, with a macro-centered sharpening.
Individual normalized-determinant residue class	Explicit small-content and all-content bounds; power saving for a polynomially growing effective modulus.
Residue-cell and determinant-frequency mean square	Unconditional bound proved here; an exceptional-frequency estimate follows.
Absolute reassembly over all windows or classes	No saving; exact partition restores the full majorant.
Signed auxiliary zero coefficient $\widehat{S}(0)$	Open; not controlled by the nonzero-frequency mean square.
Complete residual, positivity, prime-pair asymptotic, or parity breach	Not proved and not implied.

10 Exact certificate and scope ledger

The companion script

`experiments/tpc31_certificate.py`

is a standard-library Python certificate for the finite algebra, graph inequalities, and rational exponent ledger used in this paper. Run from the paper directory with Python 3.10 or later:

```
python experiments\tpc31_certificate.py
python -O experiments\tpc31_certificate.py
```

Both modes perform exactly

$$\boxed{876,320}$$

(106)

recorded checks and regenerate byte-identical JSON.

The checks are divided as follows:

Finite check family	Recorded checks
Primitive content and canonical reduced-target identities	566,738
Selected-divisor multiplier $d \mid c$ and determinant dilation	206,203
Determinant-cell degree calculations	5,939
Integer-weight Schur inequalities	78,746
Exact-content orbit occupancy	1,662
Residue partition and absolute reassembly	301
Rational splice and high- β ledgers	16,705
Scope and nonoverlap flags	26
Total	876,320

The finite identity sweep verifies

$$c = (N_m, N_n), \quad c \mid m - n, \quad (c, mn h) = 1,$$

the coprimality of the reduced targets, and $mV - nU = h\Delta^\#$. Over selected divisors it verifies $d = (r, s) \mid c$ and

$$\frac{m - n}{d} = \frac{c}{d} \Delta^\#.$$

The graph tests cover positive, negative, crossing-zero, and macro-translated windows; $q = 1$; noncoprime c, q ; content-scaled moduli cq beyond the row span (equivalently, q beyond the normalized-determinant span); full residue cells; and window-residue intersections. Exhaustive and deterministic integer weights verify the squared two-sided Schur inequality, including sharp +1 cases.

The orbit tests compare direct exact-content counts with their finite Möbius inversion and verify the one-class occupancy bound. The partition tests check that the residue cells exactly reassemble the whole determinant layer and include signed cancellation witnesses, while making no claim about the asymptotic physical coefficient. All exponent calculations use `fractions.Fraction`; no floating-point comparison is used in the high- β or threshold optimization ledger.

Reproducibility values are:

$$\begin{array}{ll} \text{normalized certificate-script source SHA-256} & 7b4c984620ce733e7741d00b3c4b547e \\ & 7d62efb05947902d91fc9dc54e133ff7, \end{array} \quad (107)$$

$$\begin{array}{ll} \text{JSON SHA-256} & 5776b4310c04df512c47f940924253d4 \\ & 75ab51a2da3c8058a52201a643479fea, \end{array} \quad (108)$$

$$\begin{array}{ll} \text{certificate digest} & cd48ae70d7d5a39b54d128a28c695557 \\ & f440de37582c7c3cec6405713eae0ab4. \end{array} \quad (109)$$

The certificate is a regression check, not a computer-assisted proof of an asymptotic estimate. In particular, it does not certify Möbius cancellation, cancellation of the auxiliary residue-cell zero coefficient, a prime-pair main term, or the existence of infinitely many prime pairs.

10.1 Claim boundary

The proved statements in this paper are exactly:

- (a) canonical reduced-target algebra;
- (b) deterministic row-graph and Schur estimates;
- (c) absolute cellwise transfer to the three retained raw channels;
- (d) the TPC-30 large-content splice;
- (e) rational exponent consequences;
- (f) residue-cell energy and exceptional-frequency compression;
- (g) the exact auxiliary residue-cell zero-coefficient and absolute-reassembly identities.

They do not imply:

- (a) cancellation of the full coprime determinant spectrum;
- (b) smallness of the auxiliary residue-cell zero coefficient in (102);
- (c) a complete Möbius-tail estimate;
- (d) a Hardy–Littlewood asymptotic or positive prime-pair lower bound;
- (e) the twin-prime conjecture or a breach of the sieve parity barrier.

The scope flags in the JSON archive encode these negative statements explicitly, making any future promotion of a finite certificate or a cellwise estimate into a global number-theoretic claim detectable in regression review.

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